KU LEUVEN BRUGGE

Analytical Evaluation of Magnetic Shielding Effectiveness of Infinite Planar Screens under Near-Field Conditions

State of the Art

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Thesis submitted for the degree of Master of Science in Engineering Technology, option Electronics and ICT

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Preface

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Albéric Simperl

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Abstract

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List of Abbreviations and Symbols

Abbreviations

EM	electromagnetic
EMC	electromagnetic compatability
SE	Shielding Effectiveness
RX	Receiving
TX	Transmitting

Symbols

$J_n(.)$	Bessel function of the first kind and order n
$\boldsymbol{B}_i$	Magnetic field in medium i
$\boldsymbol{E}_i$	Electric field in medium i
$\boldsymbol{A}_i$	First order vector potential in medium i

Chapter 1

State of the Art

1.1 What is shielding?

Electromagnetic shielding is a means of improving electromagnetic compatibility (EMC) using an electromagnetic (EM) shield. Using this shield, it is possible to reduce the electromagnetic field intensity within a prescribed region [1]. Its use cases range from military applications to civil interests. Within this broad range every application aims to achieve electric and magnetic immunity to ensure EMC.

In military settings it can be used to guarantee communication security, thus protecting sensitive information transmitted using electromagnetic waves. Implementations within the medical sector are abundant as well, this ranges from the inwards isolation of units where bio-electric signals are measured using ECG and EEG to MRI machines where an immense magnetic field is generated. Keeping this field contained within the imaging unit is necessary to prevent EM forces acting on magnetizable materials nearby. While these applications are found within controlled hospital environments, we rarely find ourselves within such environments.

Our world is filled with electromagnetic radiation and waves wherever we go. This could cause issues for people using active implanted medical devices (AIMDs), such as pacemakers and cardioverters. With the rapid rise of both electric vehicles and power electronics the risk of EM interference (EMI) follows. These vehicles produce low frequency EM fields. When such radiation can freely enter a human body, EMI might occur, hindering the performance of the sensitive detection circuits within such AIMDs. In everyday practice these effects are small to negligible, but during prolonged intense exposure the risks rise considerably. False signals could result in false detections of cardiac arrhythmia, after which the device will act upon these signals, thus giving unneeded stimulations or suppressing the pacing needed. This would then lead to arrhythmic beats. Luckily these risks have reduced to the point where driving electric vehicles shouldn't cause any problems.

Though not electronic, humans are also sensitive to these EM fields. Extremely low frequency EM fields have been classified as possibly carcinogenic to humans.

Some studies suggest a weak correlation between extended exposure to magnetic fields higher than 0.3 μT and childhood leukemia [2], [3].

In conclusion, EM shielding has become vital in the protection of our electronically dominant world. Whether it is used for the safety of sensitive electronics, isolating environments from both outwards and inward radiation or protecting transmission lines to ensure both human safety as well as a guarantee on private data transmission.

Chapter 2

The First Chapter

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2.1 Figures

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2.2 Tables

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Table 2.1: A table with the wrong layout.

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Column A	Column B	Column C
1	2	3
4	5	6

Table 2.2: A table with the correct layout.

2.3 Conclusion

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Chapter 3

Conclusion

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